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2025

# Paper To Pixel

## SMOKE TO INK CONVERSION

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- **Domain** : Smart Automation
- **Theme :** Smart Cities
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# Executive Summary

- **Context:** Air pollution from vehicles and industries releases harmful smoke. Converting this waste into usable products supports sustainability.
- **Problem Statement**: Most carbon emissions go unutilized. There’s a need for a smart system to capture and convert smoke efficiently.
- **Objective:** To develop an IoT-based system that collects smoke particles and converts them into ink, with real-time monitoring and data tracking.

# Methodology

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**Tools Used :**

1. STEP-UP TRANSFORMER – Converts 12V to 230–300V
2. VOLTAGE REGULATOR – Keeps current stable
3. CAPACITOR (1000µF) – Stores energy
4. 12V RELAY – Controls power switching
5. 12V GEL BATTERY – Main power source
6. ARDUINO UNO – System controller
7. SMOKE SENSOR – Detects smoke
8. BUZZER – Alerts when smoke is detected
9. COLLECTOR (FILTER) – Captures carbon
10. STEP-UP IN COLLECTOR – Boosts power for filtering

# Methodology

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```
+-------------------+     +----------------------+     +------------------------+
|                   |     |                      |     |                        |
| Pollution Source  +---->+ Pollution Collection +---->+ Soot Extraction &      |
| (e.g., vehicle    |     | Unit (e.g., Kaalink) |     | Purification           |
| exhaust, factory  |     |                      |     |                        |
| emissions)        |     |                      |     |                        |
|                   |     |                      |     |                        |
+-------------------+     +----------------------+     +------------------------+
                                   |                          |
                                   v                          v
                        +-------------------+     +----------------------+
                        |                   |     |                      |
                        | Carbon Purification|     | Pigment Processing   |
                        | & Detoxification  |     | (Grinding & Mixing)  |
                        |                   |     |                      |
                        +-------------------+     +----------------------+
                                   |                          |
                                   v                          v
                        +-------------------+     +----------------------+
                        |                   |     |                      |
                        | Ink Formulation   |     | Quality Control &    |
                        | (Binder & Solvent)|     | Packaging            |
                        |                   |     |                      |
                        +-------------------+     +----------------------+
```

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Block Diagram of Smoke to Ink Conversion

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# Methodology

The process begins with

- Step 1 : Where a step-up transformer is used to convert 12V DC to 230V AC, enabling the system to operate with standard voltage.
- Step 2 : The smoke collector unit is activated, which captures carbon particles from the emissions.
- Step 3 : Involves powering the system by turning on the battery, ensuring uninterrupted operation.
- Step 4 : The Arduino microcontroller and IR sensor are activated, enabling real-time detection, data processing, and monitoring of the smoke collection process.

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# Innovation, Analysis & Results

Working Prototype Model of
Smoke to Ink Conversion

Results :

- Successfully captured carbon from smoke
- Converted captured particles into black ink
- Proved that smoke can be reused instead of wasted
- System worked effectively for real-time monitoring

**Innovation :**

- Converts smoke into usable ink
- Promotes zero-waste and eco-friendly tech
- Low-cost and compact design for easy use

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# Conclusion & Future Work

**Real-World Applications**

- **Factories & Industries – Capture smoke and turn waste into ink**
- **Vehicles (with exhaust filter) – Reduce air pollution on roads**
- **Eco-Friendly Printing – Use recycled ink for sustainable printing**

**Limitations**

✓ Limited Smoke Capture – Not suitable for open or windy environments

✓ Power Dependency – Needs stable battery or external power supply

✓ Maintenance Needed – Filters and sensors require regular cleaning or replacement

# Conclusion & Future Work

**Summary of Contribution**

❖ Developed an IoT-based system to convert smoke into ink

❖ Reduced air pollution by capturing carbon particles

❖ Promoted sustainable waste-to-resource technology

**Future Enhancement**

❖ Add solar power for eco-friendly energy

❖ Improve ink quality for commercial use

❖ Scale up for use in large industries